Personal Data Analytics Project

Credit Risk Modelling in Python

Application of Statistical Models to Expected Credit Loss Estimation

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Personal project completed from June 2026 to September 2026

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Python · Credit Risk · Statistical Modelling · Data Analysis

Project completed as part of a guided Udemy course

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Executive Summary

This report presents an end-to-end credit risk modelling exercise based on historical Lending Club loans. The workflow connects data preparation, a probability-of-default model and scorecard, model monitoring, recovery and exposure modelling, and a portfolio-level Expected Loss calculation. Its purpose is to demonstrate how different statistical estimates can be combined into a coherent credit-risk analysis, rather than to deliver a production-ready regulatory model.

The final portfolio contains 466,285 loans and approximately $6.664 billion in funded amounts. On 93,257 holdout observations, the PD model achieved an AUROC of 70.18%, a Gini coefficient of 40.35% and a KS statistic of 29.68%. The integrated Expected Loss was approximately $512.44 million, or 7.69% of the funded amount. These figures describe the reproduced model and dataset; they are not forecasts for a current lending portfolio.

1. Project Background and Objectives

A lender's potential loss depends on more than whether a borrower defaults. The severity of a default and the amount outstanding at that point also matter. Credit risk modelling therefore distinguishes Probability of Default (PD), Loss Given Default (LGD) and Exposure at Default (EAD). Multiplying these quantities produces an expected loss estimate for each loan, which can then be aggregated across a portfolio.

Expected Loss = PD × LGD × EAD

The work follows a guided educational methodology and is conceptually related to the PD/LGD/EAD frameworks used in credit risk. It does not implement all requirements of Basel regulatory capital or IFRS 9 impairment accounting, such as comprehensive lifetime default projections, forward-looking economic scenarios, staging, discounting or independent model governance.

2. Data Understanding and Preparation

The development dataset covers 466,285 historical Lending Club loans from the 2007–2014 period. Borrower information, loan characteristics, credit-history variables, repayment status and recovery information provide the inputs for the models. For recovery and exposure modelling, 43,236 charged-off loans were isolated because their realised outcomes can be used to define the corresponding targets.

Preparing the data was a substantial part of the exercise. Employment length, loan term and date fields needed to be standardised; numerical characteristics were grouped into interpretable intervals; categorical values were converted into dummy variables. The PD model used grouped variables and Weight of Evidence (WoE) analysis to examine how each category relates to the observed good/bad outcome. One reference category per grouped characteristic was excluded from the regression to avoid redundancy.

Reproducing the notebook with newer libraries revealed issues that were not immediately apparent from the original code. In particular, the issue-date calculation had to reference the actual issue-date field, employment lengths below one year needed explicit handling, and unsupported month-

based NumPy date arithmetic had to be replaced. Missing-value treatment also had to be applied consistently when LGD/EAD models were later scored on the complete portfolio. These corrections mattered because inconsistent preprocessing can silently alter a model's input variables and ultimately its predictions.

3. Probability of Default and Credit Scorecard

3.1 Logistic regression and model performance

The PD model uses logistic regression on WoE-informed, grouped borrower and loan characteristics. Its output is a probability of default, with the model's target coding taken into account when selecting the relevant class probability. Approximately 80% of observations were used for training and 20% for holdout evaluation.

On the 93,257-loan test sample, the AUROC reached 70.18%. In practical terms, this measures how well the model ranks borrowers with different observed outcomes, across classification thresholds. The corresponding Gini coefficient was 40.35%, while the KS statistic was 29.68%. These results suggest useful, but far from perfect, discrimination. They should not be mistaken for evidence that individual default probabilities are necessarily well calibrated.

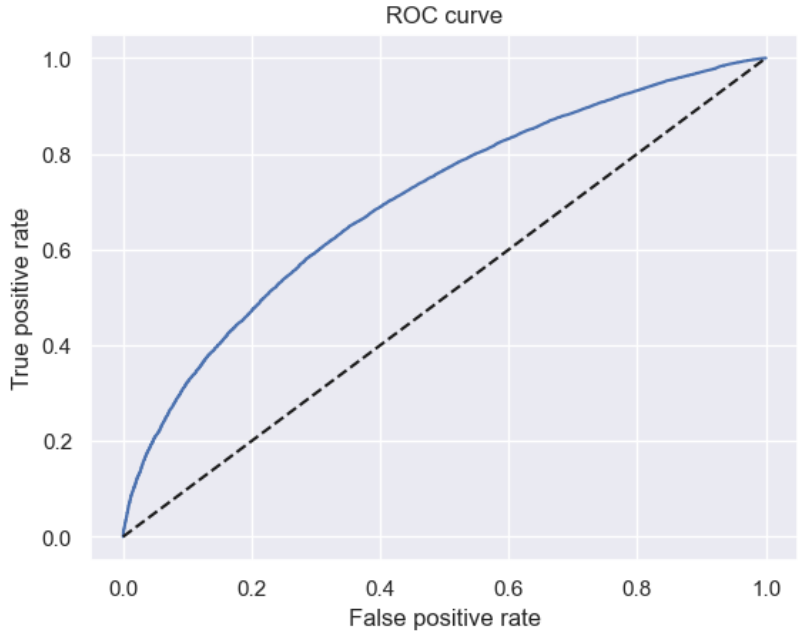


Figure 1. ROC curve produced by the PD modelling notebook.

Accuracy was approximately 58.60% at the threshold used in the notebook. This is not the main performance indicator here: classification accuracy depends on the selected cut-off and can be misleading when the classes are imbalanced. Ranking measures such as AUROC are more informative for assessing this particular credit scoring exercise.

3.2 From probabilities to scores

The logistic model was translated into a points-based credit scorecard. Each grouped characteristic contributes a number of points derived from the fitted coefficients, and the points are summed to obtain a borrower score. This makes the model easier to communicate and allows score thresholds to be related to potential approval and rejection decisions.

The scorecard is an illustration of how an interpretable statistical model could be operationalised. Choosing a real approval threshold would require additional business information, including risk appetite, expected profitability, customer outcomes and calibration checks.

4. Monitoring the Scorecard

A model's development performance is only part of the picture. The monitoring notebook applies the scorecard to a separate 2015 population and uses the Population Stability Index (PSI) to compare score and characteristic distributions with the development data. PSI measures distributional change, not directly predictive accuracy or calibration.

The overall score PSI was 0.0873, which is relatively limited under the reference convention used in the project. However, the feature-level results were more mixed: initial listing status reached 0.3337 and months since issue date 0.3048. Those shifts deserve closer investigation even when the aggregate score distribution appears reasonably stable.

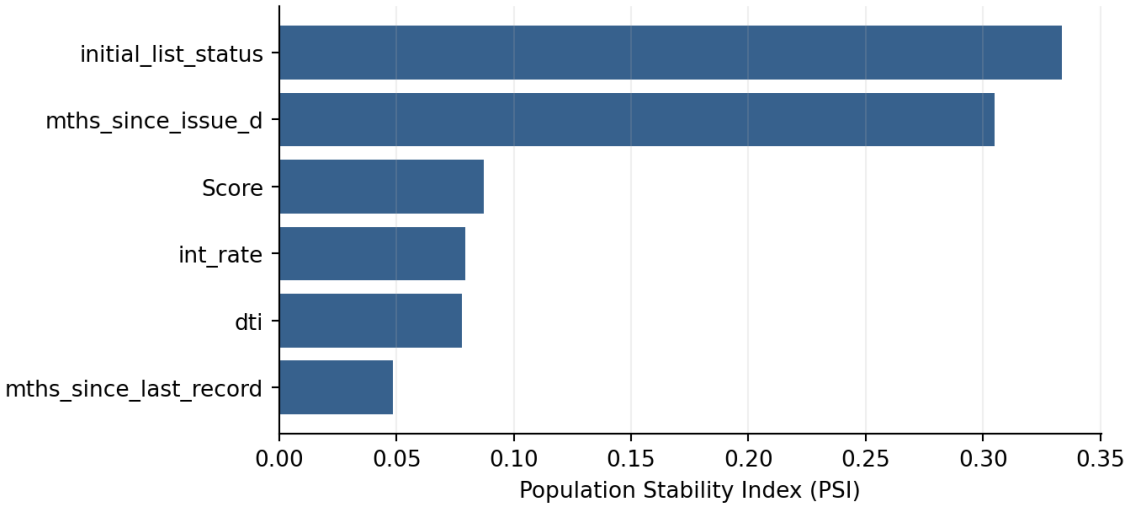


Figure 2. Selected characteristic-level PSI results and overall score PSI.

This contrast is one of the most useful findings from the monitoring exercise. A stable overall score can conceal important changes in individual inputs. In a production setting, these changes would prompt checks of population composition, variable definitions, data pipelines and subsequent model performance. PSI thresholds are operational conventions, not universal regulatory benchmarks.

5. Loss Given Default

LGD estimates the proportion of exposure not recovered after default. The modelling dataset therefore focuses on charged-off loans and uses observed recoveries to construct a recovery-rate target. The course methodology separates the problem into two stages because some loans have no recorded recovery at all, while others have a positive but variable recovery rate.

$$\textbf{LGD} = 1 - \textbf{Recovery Rate}$$

Stage 1 is a logistic classification model identifying whether recovery is positive. It achieved approximately 63.10% test accuracy and an AUROC of 66.54%. Stage 2 is a linear regression estimating recovery severity, with a test correlation between observed and predicted recovery rates of about 0.308. This relatively weak relationship highlights the difficulty of predicting recovery outcomes at the individual-loan level.

An important implementation detail is that the notebook multiplies the Stage 1 predicted binary class (0 or 1) by the Stage 2 predicted recovery rate, rather than multiplying the probability of recovery by conditional recovery severity. The combined prediction is clipped to the interval from zero to one before calculating LGD. Across the scored portfolio, the average predicted LGD was 94.23%. That high value reflects the model's recovery assumptions and should be interpreted in the context of this historical dataset and educational implementation.

6. Exposure at Default

EAD represents the amount expected to remain exposed when default occurs. In this course, an intermediate variable called the Credit Conversion Factor (CCF) is constructed from the funded amount and principal already repaid. For amortising loans, it functions as a remaining-principal factor rather than the conventional regulatory conversion factor used for undrawn commitments.

Project CCF = (Funded Amount − Principal Repaid) / Funded Amount

EAD = Funded Amount × Predicted CCF

A linear regression was fitted to estimate this factor. The test correlation between actual and predicted CCF was approximately 0.531, indicating a moderate relationship but substantial remaining unexplained variation. Predictions were bounded to the interval from zero to one before calculating EAD. The resulting average predicted exposure across the portfolio was $10,814.85.

The distinction between the course's CCF definition and the conventional off-balance-sheet CCF matters: the terminology is retained to stay faithful to the notebook, but the estimated quantity is specifically a share of principal remaining outstanding.

7. Portfolio Expected Loss

The final step scores the full portfolio with the PD, LGD and EAD models and multiplies the three estimates loan by loan. Summing those individual estimates produces a portfolio Expected Loss. The denominator used to calculate the reported loss rate is the total funded amount, rather than total EAD or a regulatory capital measure.

$$\text{Total Expected Loss} = \Sigma_i \, (PD_i \times LGD_i \times EAD_i)$$

Portfolio indicator	Final rerun result
Loans assessed	466,285
Average PD / LGD	10.93% / 94.23%
Average EAD	$10,814.85
Total funded amount	$6,664,052,450
Total Expected Loss	$512,436,986.69
Expected Loss / funded amount	7.69%
Mean / median EL per loan	$1,098.98 / $722.97

The portfolio-level Expected Loss of $512.44 million is approximately 7.69% of the $6.664 billion funded amount. The mean individual Expected Loss ($1,098.98) is noticeably above the median ($722.97), consistent with a right-skewed distribution in which a smaller group of loans contributes disproportionately to estimated losses. The maximum individual estimate was approximately $11,720.

These outputs illustrate how risk can accumulate across a portfolio even when many individual exposures are relatively small. The total should not be compared directly with regulatory capital ratios: expected credit loss and capital requirements measure different concepts and are calculated under different frameworks.

8. Practical Implementation and Learning

Although the statistical design followed a guided Udemy course, completing the project required understanding how each notebook's outputs fed into the next stage. The most challenging aspect of the reproduction was maintaining consistent feature definitions across data preparation, PD estimation, monitoring and portfolio scoring while working with a newer Python and scikit-learn environment.

The adjustments included modernising custom regression wrappers, addressing deprecated APIs, correcting a date-field mismatch, making the scorecard feature ordering robust, and resolving missing values that appeared when the recovery models were applied to the complete dataset. These tasks provided useful experience with debugging, data quality and the difference between code that executes successfully and a modelling pipeline that remains methodologically consistent.

9. Limitations and Interpretation

The analysis is an educational reproduction using a historical, platform-specific loan population. The results cannot be assumed to generalise to present-day lending conditions. Moreover, a model can discriminate between borrowers without producing well-calibrated default probabilities; further validation would be required to assess calibration, performance across time, and robustness to shifts in the underlying borrower population.

The LGD combination uses a hard Stage 1 classification rather than a probability-weighted expectation, and the EAD factor is not the standard regulatory off-balance-sheet CCF. Monitoring through PSI identifies distributional changes but cannot establish ongoing predictive validity on its own. A full accounting or regulatory deployment would additionally require governance, independent validation, scenario analysis and other framework-specific components.

10. Conclusion

I undertook this project to build a practical understanding of how statistical models can be used to quantify credit risk. Working through the guided methodology helped me connect data preparation and feature engineering to model evaluation, scorecard construction, monitoring and expected loss estimation. Adapting the notebooks to a current Python environment also reinforced the importance of reproducibility and careful debugging.

The final results—70.18% PD AUROC, 40.35% Gini, and $512.44 million in estimated portfolio Expected Loss—summarise the outcome of the reproduced pipeline. More importantly, the project provided a structured opportunity to understand the role, limitations and interactions of PD, LGD

and EAD models, and to practise communicating quantitative results without overstating what they demonstrate.

Appendix — Selected Model Diagnostics

Component	Metric	Result
PD	Holdout AUROC / Gini / KS	70.18% / 40.35% / 29.68%
Monitoring	Overall score PSI	0.0873
LGD Stage 1	Accuracy / AUROC	63.10% / 66.54%
LGD Stage 2	Actual–predicted correlation	0.308
EAD	Actual–predicted CCF correlation	0.531

Source: executed project notebooks (Data Preparation, PD Model, Monitoring, and Calculating Expected Loss). Metrics are reported as displayed in the rerun outputs and rounded where appropriate.